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Summary

AI Engineer with hands-on production experience designing and shipping multi-agent AI systems using **LangChain**, **LangGraph**, and **LangSmith**. Proven track record building RAG pipelines, autonomous agent workflows, and REST APIs deployed at scale with **FastAPI** and **Docker**.

Experience

Python AI Engineer at Arcitech.ai, Navi Mumbai Feb 2026 – Present

- Designed and shipped **multi-agent AI systems** in production using **LangChain**, **LangGraph**, and **LangSmith** for orchestration, real-time monitoring, and latency optimization across concurrent agent workflows.
- Designed **clickable AI suggestion cards** as a first-class response pattern, significantly improving user discoverability and engagement with agentic outputs.
- Built **LLM-ready web-scraping data pipelines** replacing unreliable PDFs, creating a scalable foundation for advanced RAG; proactively audited APIs and resolved production data-quality issues end-to-end.
- Drove **GPT model upgrade** via comparative benchmark analysis; re-engineered image upscaling pipeline, resolving hallucination defects and delivering native 2K/4K output quality in production.

Python AI Intern at Arcitech.ai, Navi Mumbai Dec 2025 – Feb 2026

- Built **ADFactors**, a multi-agent PR generation system using **LangChain** and **LangGraph**; integrated compliance agents, **web search**, and PDF generation with cross-functional team collaboration.
- Monitored agent performance with **LangSmith**; reduced pipeline latency by replacing high-latency reasoning models with task-optimised alternatives, improving end-to-end response throughput.
- Developed **REST APIs** with **FastAPI**, **PostgreSQL**, and **pgvector**; built **RAG systems** with persistent memory, enabling context-aware retrieval over structured and unstructured data sources.

AI/ML Intern at Annam.ai, IIT Ropar – Center of Excellence in Digital Agriculture Aug 2025 – Oct 2025

- Applied **MLOps** practices with **MLflow** for experiment tracking and model versioning; deployed models via **FastAPI** and **Docker**; explored **LLMs** for intelligent automation in a research setting.
- Conducted computer vision research using **PyTorch** and **Transformers** for high-performance agricultural image modeling.

Projects

RAG Pipeline for Hotels github.com/romit-23/RAG-Model-for-Hotels

- Built an end-to-end **RAG pipeline** for hotel booking insights using **FAISS**, **Mistral-7B** (Hugging Face), and **LangChain**; deployed with **FastAPI** and **Docker** as a production-grade API.
- Performed comprehensive **data cleaning and EDA** to improve retrieval quality and downstream LLM response accuracy.
- Tools:** LangChain, FAISS, Mistral-7B, FastAPI, Docker | **Keywords:** RAG, Semantic Search, Vector Databases

Financial Fraud Detection System github.com/romit-23/Fraud-Detection-Model

- Built ML model on 6.3M transactions (0.13% fraud rate) using **Random Forest** & **XGBoost** with **SMOTE** for class imbalance; engineered domain-specific features to improve precision and recall.
- Evaluated models using **AUC-ROC** and **F1** metrics on high-volume structured transaction data.

GuardNet: Violence Detection in Public Surveillance github.com/romit-23/GuardNet

- Hybrid deep learning model (ConvNeXt-Tiny + LSTM + attention) for real-time CCTV violence detection; [published at IEEE](#).
- Tools:** PyTorch, ConvNeXt, LSTM, OpenCV, CUDA | **Keywords:** Computer Vision, Temporal Modeling, Deep Learning

Technologies

Languages: Python, JavaScript, SQL, Bash, HTML

Agentic AI & Orchestration: LangChain, LangGraph, LangSmith, Multi-Agent Systems, RAG, Semantic Search, Chatbots, NER

LLMs: GPT-4, Llama-3, Claude 3, Mistral-7B | Generative AI, Prompt Engineering, Retrieval-Augmented Generation

Tools & Platforms: FastAPI, Docker, AWS, Git/GitHub, Django | **Databases:** PostgreSQL, pgvector, MySQL, FAISS

ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, SpaCy, NLTK, NumPy, Pandas | **MLOps:** MLflow

Data Viz: Matplotlib, Seaborn, Plotly, Tableau

Education

SIES Graduate School of Technology , B.Tech in Artificial Intelligence And Machine Learning	2021 – 2025
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